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Framework to Solving Product Sense Interview Questions for Data Science

How to Prepare for Product Sense Interview Questions before Entering the Data Science Interview

When going to a technical interview, most companies will ask product questions based on current/potential complications involving or related to their product. This could range from to some degree a straightforward question about their product to broad questions on how much you understand the product.

For example, [Quora asked](#), "What metric would you use to measure the impact of a search toolbar change?" This is a more straightforward question, where possible solutions could be CTR (Click through rate) and A/B testing. A more vague question for example [Uber has asked](#) "Pick your favorite product or app and describe how you would improve it or design it." This is a type of question to see how much you analyze products and for the interviewer to understand your thought process.

For the first example question, there were more defined answers an interviewer would like to see, where the second question is a more general question to see why you chose certain features to improve over others and understand how that thought process could be applied to their company's product.

Before entering the interview, research the company's product. Since product based questions are referencing the company's product, understand the company's product. What are its competitors (if any) and common challenges faced? While it is impractical to analyze every company's product thoroughly, especially for multinational corporations, it is recommended to have an overall understanding of the company's products. When companies have multiple products, they usually offer Data Science jobs under specific divisions when applying. For example, Google has multiple divisions that require data scientists and you have applied for "Marketing Automation, Google Ads". Look into what this product is about. Companies want to improve their product and if you understand their product and have feasible solutions to their product, this makes you more likely to be hired.

If a specific division is not included in the job listing, you can move on to the next step of checking the job responsibilities. The job responsibilities are meant for you to get a better understanding of what the job entails and potential questions from the interviewer. Glassdoor provides a [description](#) given by most data science jobs with the following responsibilities.

- Work with stakeholders throughout the organization to identify opportunities for leveraging company data to drive business solutions.
- Mine and analyze data from company databases to drive **optimization and improvement of product development, marketing techniques, and business strategies**.
- **Assess the effectiveness and accuracy of new data sources and data gathering techniques.**

- Develop custom data models and algorithms to apply to data sets.
- Use predictive modeling to increase and **optimize customer experiences, revenue generation, ad targeting, and other business outcomes.**
- Coordinate with different functional teams to **implement models** and **monitor outcomes.**
- Develop processes and tools to **monitor and analyze model performance and data accuracy.**
- **Design data modeling processes** to create algorithms and predictive models and perform custom analysis
- Communicate findings to stakeholders **using visualization** and other means
- Develop **forecasting models**
- Publish results and **address constraints/limitations** with business partners.

The highlighted words are all terms to focus on. Generally, most of these terms are seen in many job descriptions, so it is recommended to practice on questions involving these topics.

Once you get the list of important keywords from the job ad, analyze what type of techniques could be applied for each item on the list. It should be noted as well when a company mentions vaguely about a certain responsibility, such as in the above case “other business outcomes”, you should not completely ignore that, but just have a general list of techniques that are commonly used around the industry, such as PCA (Principal Component Analysis).

It would be helpful for you to create a personal document that has a list of common terms from job descriptions and algorithms/concepts you know. Classify the algorithms/concepts under the common terms in job descriptions. For example, Ad targeting is a common topic asked. Under Ad targeting, you could include CTR, Total site traffic, Page Value. These 3 concepts are all linked with Ad Targeting and can be used when discussing Ad Targeting related questions. Create multiple lists like this and refresh your memory on the most applicable lists for Company X before the interview.

Sometimes specifics may be asked, such as PayPal had a Venmo Risk Data Scientist job description that included “Produce accurate, automated risk solutions, and manual handling policies to block fraudulent activities by declining payments, adding accounts to the fraud review queue, or restricting suspicious users.” This entire line is specific and generally not seen in other data science job descriptions. It is understood that risk solutions are commonly involved in this job. Think about the common problems that risk solutions companies face, or even better what risk-related problems does PayPal face. This allows you to research and understand more about the company and **what type of problems your potential employer faces.** These common problems will be the type of questions you are asked during an interview. Think about what type of product sense interview questions companies will ask related to their product from a data science/product manager perspective.

Approaching Product Sense Interview Questions



Product sense interview questions in data science interviews are not only asked for getting the most accurate answer possible. It is more for the interviewer to understand your approach to your answer. These questions are asked for the following reasons:

- Diagnose and solve real product problems
 - Possibly see how familiar you are with the company's product/business
- How practical your solution to a problem is
- Evaluate Candidates' ability to define metrics
- Understand the impact and tradeoffs in the metrics measured
- Communicating your solution in an effective and structured manner

(These are all directly tied into how a practicing data scientist will explain the reasoning behind a problem and proposed solutions. Keep this in mind while answering questions)

Before trying to give an immediate response to the question, first, you must understand the type of product interview question that is being asked. There are **3 types** of product sense interview questions that are asked during interviews.

1. Analyzing a metric related problem (Most common)
2. Measuring impact of a new product/feature
3. Designing a product

All three of these product sense interview questions will require a different type of approach to the answer but are always tied to **revenue, market share, or user engagement**. Understand if the question is related to which of the three (or more than one) and structure your solution accordingly.

Analyzing a metric related problem

This type of product sense interview question usually mentions a metric related to a company's product and mentions it has been negatively affected. Your solution should give your thought process on why this has occurred.

Examples of these types of product sense interview questions:

1. Feature X has been decreased in use by N%
Uber asks "Uber Black rides have dropped 10%. How would you investigate this reduction?"
2. Feature X is down by N% but Feature Y is up by M%
Google asks "There is an increase of users who feel their privacy is respected by 8%, but the engagement to the Google Play Services has decreased by 14%. How would you investigate this discrepancy?"

Companies look for:

- How well the interviewee can define and explain the relevance of metrics to the solution
- Understand what could affect the success of certain features/products
- If there is a metric-tradeoff, which metric matters more according to the situation

Another metrics impact question companies might ask is how certain factors might affect the metrics measured. Uber has asked "Explain how network effects might influence your choice of how to assign experimental/control units and measure your main outcome metrics" This type of question tests how well you understand what could affect the feature/product but also to what extent.

Measuring impact

This type of question asks you to measure how effective a certain feature/product of a company is.

Examples of these types of product sense interview questions:

1. Quora asked, "What metric would you use to measure the impact of a search toolbar change?"
2. YouTube asks "How would you measure the success of YouTube shorts?"

Companies look for:

- How well the interviewee can define and explain the relevance of metrics to the solution
- Understand the purpose and what factors could affect the success of a feature/product
- Design an experiment to measure how successful the feature/product is based on the defined metrics

Designing a product

These are completely open-ended questions where there is no right answer. These are usually the hardest questions in your Data Science Product interviews since these test your thought process rather than your answer. These questions could be directly related to a company's product or a more general question that would be related to the company's product!

Examples of these types of questions:

1. **Lyft asked** "Describe how to engineer the heatmap telling taxi drivers where to go to maximize their probability to get a client. How do you define which area will have high demand next and who do you want to go there?"
2. **Microsoft asked**, "What would you do to summarize a Twitter feed?"
3. **Yelp asked**, "If you had to propose a new Yelp feature, what would it be?"

This type of product sense interview question sometimes includes how to measure the success of your designed solution, for example when you mention your new Yelp feature, Yelp may ask how you will determine if this product is successful to launch? This will be tied back into the measuring success type of questions.

Companies look for:

- How well you understand the company's products/market
- How well you understand the process of developing a product (Target audience in mind)
- How well you can define and explain the relevance of metrics to the solution
- How well you understand how to measure the success of the product

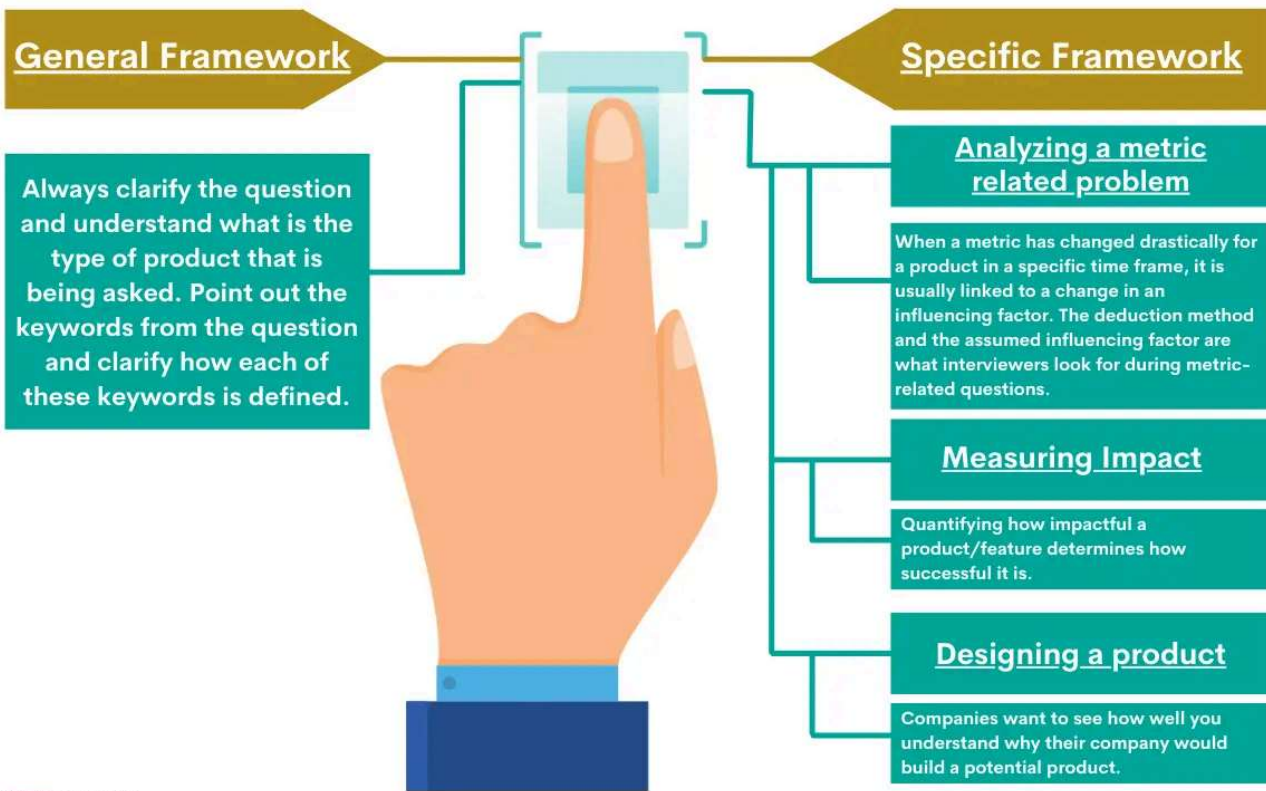
Structuring Solution

Once you've understood the type of product sense interview question the interviewer is asking. You have moved on to the most important step, the structure of your solution. Interviewers do not look just for a possible answer, but for multiple factors.

The interviewee must show that they:

- Have a systematic approach to your answer
- Cover all important aspects of the question
- Ensure the practicality of the solution

Structuring Solution



Although each type of product sense question has a different framework for each solution, there is a general framework that is followed by all types of product questions. The general framework is the basis of your solution since it shows that you have an understanding of the product and things to look out for.

General Framework

The interviewee should **always clarify the question**. As direct as the question may seem, you should always ask questions about the keywords in the question. Take some time to understand the question and its metrics and goals.

You must define the metrics that are being used in the product interview question. Metrics that are mentioned in the question and metrics that you are bringing in to help form your solution should not only be defined but an **explanation of the relevance of the metric to the goal** should be stated. When you explain why a metric is used, it shows that you as the interviewee understand why a certain metric is relevant to the question and not a random guess. Discuss with your interviewer about the goals/metrics until you and the interviewer are on the same page about these. Remember that the rest of your answer is based on these metrics/goals, so if your interviewer does not know why you are choosing a certain metric or see any correlation between the metric and the goal of the product, this would negatively affect your chances of getting hired.

You should also understand **what is the type of product that is being asked**. Is your question based on a **physical/digital product or a feature of a product**?

- Physical/digital product: Clarify with the interviewer if you need to answer based on the product any specific part/feature of the product? Eg: **Yelp has asked**, "If you had to propose a new Yelp feature, what would it be?" Yelp has multiple products under it, such as business reviews or table reservations. Ask the interviewer Is there any specific part of the product to improve or the product as a whole?
- Feature of a product: Make sure you understand how the feature works. You should chew the question to the bone. Ask the interviewer if there's any part of the feature you do not understand or would like clarified. If the interview does not know or tell you, make assumptions about the feature and ask the interviewer if you can make those assumptions.

After understanding what the product/feature is, **point out the keywords from the question and clarify how each of these keywords is defined**. It is recommended to define the keywords in the question from start to end, so you cover every important element.

Example: **eBay asked** "eBay has to identify the cameras from the other objects like tripods, cables, and batteries. What would be your approach? Data includes ads title, description of the product, price, images, etc."

- "Camera" → Does the camera contain all types of cameras or any specific camera such as point-shoot cameras/video cameras/action cameras?
- "Other objects" → The examples are given are all linked to cameras, so are we trying to differentiate cameras from objects that are related to a camera or any object in general?
- "Data" → How often was this data collected? This matters because technology changes every decade, so a more accurate classification algorithm will recognize modern cameras better, which are more likely to be sold on eBay than cameras from the 1990s. Another question about data is data accuracy. How accurate is the data of what a camera is?
- "Ads title" → Cameras are often in bundles, such as with tripods/cables/batteries (specifically what the camera needs to be differentiated from) so will ad titles contain information only related to the camera or will it include other products in the title as well?
- "Description" → Does the description indicate the verified specifications about the product produced by the company or a user-written description of what the product is? Does the description include all the items in the product image? Since some images contain the entire bundle, the classification algorithm must be able to detect a camera and ignore the tripods and others.
- "Price" → Is price measured by the MSRP of the product, current auction price, buy-it-now price, or any other measuring technique? [This is a part that shows how well you understand the company's product since most other retailers won't have an auction price]
- "Images" → Are these images user-submitted or the professional images that were released by the manufacturer?

When related to a product, **time is always an influencing factor** whether a metric is affected due to the time of year or the goal needs to be met before a certain timeframe. Time influences people's decisions in purchasing a feature/product from a company. When a metric has noticed a change, define when the change starts-ends also possible if the metric change repeats during a certain time frame each year. You should also ask whether the metric change is a constant change or a sharp change during a certain timeframe? For example, the number of purchases may be going up during national holidays due to the company offering discounts.

A product is affected by **internal/external factors that influence the metrics/goals**. Mention the important internal and external factors (mention at least one) that could affect the metrics/goals and explicitly state if the factor is internal or external.

Examples of internal/external factors:

1. Internal Factors:

1. **Data accuracy** - How accurate is the metric measured / data collected?
2. **Data volume size** - Is there enough usable data that could be used to produce accurate models?

3. Product quality - Do customers, not like the product? Has there been a change in the product quality, due to changes in the cost of the product or new organization changes?

2. External Factors:

1. User demographics - Are the major changes in the metric coming for a certain demographic of users? Example: A decrease in Facebook user creations in the past year could be linked to younger people prefer to use Instagram/Snapchat compared to Facebook as a social media platform.
2. Technological Problems - Any change in technology demand/supply required to produce the product or is the product/feature not functioning properly due to a major bug? Example: Sony Playstation 5 is in shortage due to the shortage of Intel processor chips.
3. Social Shift - A switch in social taste is a major shift that could affect the entire product. Example: Blockbuster had to close down since people's social taste shifted to stream movies online using Netflix.
4. Complementary Goods - If the number of sales of a complementary product falls, the sales of the product in question will fall as well. Example: Apple noticed a decrease in the number of people buying AppleCare last year, this could be due to a significant decrease in the number of iPhones sold last year.

All of the above is just meant for clarifying the question and to show that you understand the important parts of the development of a product and what are some basic factors that could affect it. At this point, you should refer to the specific questions framework section below according to the type of question.

Specific Framework

Analyzing a metric related problem

When a metric has changed drastically for a product in a specific time frame, it is usually linked to a change in an influencing factor. The deduction method and the assumed influencing factor are what interviewers look for during metric-related questions.

The first check would be to see if any other products/features from the company have been affected. Name other products/features from the company and mention if there have been similar effects on other company's products it may be an internal factor such as organizational change or external factor such as the company reputation has been damaged.

If any other company's products have not been affected, the second check would be to see if any competition's products have been affected. If their products have also been affected similarly to the company's product, it can be deduced that it is an external factor such as shortage in parts for the product or shift in social tastes.

If only the product has been affected and none of the competition or other company's products, understanding what factors/metrics directly influence the metric is next. For example, Twitch has found that the daily active users have fallen by 5% after weeks of increasing daily active users. A directly influencing factor could be the total site traffic has been overloaded causing the website to have long loading times. This could be due to the web servers not being able to handle the large number of users trying to access the website. Here the total site traffic was a metric that negatively affected the number of daily active users.

This ties directly into the concept of pain points. Pain points are parts of the product that displeases the customer that may voluntarily stop them from using it. (In the previous example, slow website load time would be the pain point) List a couple of potential pain points that would make users dislike the product. Try to order the pain points from most relevant to least relevant, so it's helpful for the interviewer to understand how well you understand what is relevant to the product. After mentioning pain points, mention how your solution will solve the users' pain points.

In reference to the Twitch situation mentioned previously, Twitch is currently trying to increase the number of web servers to decrease website load time which in turn increases the daily active users. This is a straightforward approach, but what happens when a **tradeoff is involved?** For example, Microsoft asks "During the pandemic, it was predicted that there will be an increase in Skype daily active users, but there was a decrease by 4%. Why do you believe this occurred?" A reason could be due to Microsoft maintaining **both Skype and Teams**, which both are forms of video communication. With the Covid-19 pandemic, Microsoft pushed Teams to the public's eye more than Skype. This would result in a decrease in Daily Active Users in Skype but an increase in Teams. Since both are Microsoft products, Microsoft has to make a decision of this tradeoff, whether to accept the decrease in DAU in Skype and an increase in Teams. These are important decisions you have to make during interviews with a strong reason. As the interviewee, you must decide which aspect of the tradeoff is more important and provide a well-thought-out solution. A general rule of thumb is, when the metric of the tradeoff is involved in the question, favor the metric at hand. For the Microsoft-Skype example, you should favor Skype DAU in this tradeoff.

Measuring Impact

Quantifying how impactful a product/feature determines how successful it is. When measuring impact, you should **define 3 metrics: 2 metrics that measure success and 1 metric that does not worsen when the new feature/product**. Make sure when mentioning the 3 metrics: define the metric, explain the relevance of the metric to your goal, and possibly why you chose certain metrics over another.

Measuring success is obviously an important part of understanding how well a product/feature is performing. Understand **what metrics would prove the product has been successful**. Common examples are: **Organic growth of users** has increased, profits have increased, tweets about the new product/feature are positive.

Interviewees always mention how to measure success, but often forget that making sure certain metrics improve is not always a positive sign. Most metrics should not worsen in the path of a new product. For example, YouTube released a new feature in its mobile app called shorts, which is similar to the TikTok format. This was released to increase Daily Active Users of the target audience of younger users, who prefer the TikTok format. While Daily Active Users and Organic growth may increase, YouTube should make sure the user engagement rate does not decrease.

Since the goal of this type of product sense interview question is to measure impact, for each of these metrics you should give a **quantifiable number of when this metric can be deemed as a positive**. This should show how well you understand when a product can be deemed successful. For example: **If at least 80% of tweets are positive about the new products, customer satisfaction with the product is positive**. If organic growth has increased by **8% that is a positive**. If the user retention rate does not decrease by more than 5%, that is a positive. Do not provide a metric value that is extremely easy/hard to achieve. For example: Do not mention that only if 99% of tweets are positive, customer satisfaction with the product is positive. 99% of users liking a product is an unrealistic goal.

After mentioning, defining, and explaining your reason for choosing these metrics for the question, you need to explain what methodology you are going to use for measuring these metrics. **How will you derive the metric?** For example, Tweets about the product that are positive can be derived by **running sentiment analysis on all public tweets and replies that mention the product to deduce if people like or dislike the product**. Simple metrics such as **Daily Active Users** usually do not need a derivation.

There is a general experiment to test the success of a new feature **A/B testing** which is similar to the concept of control and treatment groups.

When explaining any experiment, mention what user group you are testing on. Are you trying to expand the product to new user groups? You would want to run your experiment on specific demographics. Are you trying to measure how the general public reacts to the new product? You would want to run the experiment on a random group.

Designing a product

Companies want to see how well you understand why their company would build a potential product. Why is the company building this type of product a great help to understand why a product is being developed, who is the target audience, and advertising the product? Mention why a company would be interested in building a potential product by connecting what type of products the company produces. Lyft asks "Describe how to engineer the heatmap telling taxi drivers where to go to maximize their probability to get a client. How do you define which area will have high demand next and who do you want to go there?" This new feature could be helpful for Lyft and its drivers for connecting drivers and riders faster, thus generating more revenue for Lyft and its drivers.

When a product needs to be designed first understand the target audience. Remember the target audience when designing your product. If the question nor the interview specify the target audience, state who you believe is the target audience and mention you are creating a product based on this target audience in mind. Continuing with the Lyft example, the target audience is mentioned as taxi drivers, but this most likely refers to the Lyft drivers. (These are assumptions you should ask your interviewer if this is a valid assumption!)

Understand what are the goals and KPI (Key Performance Indicator) in the product. KPIs are the key metric in a business problem, so this would dictate what metrics should be measured and improved. KPI is based around these two types: Product/System Performance and User Experience. Examples of Product/System performance: App initialization time, Time taken to connect to the server. Examples of User Experience: Page Value, Exit Rate. Remember your KPI should be the main focus on achieving your goal when designing your product.

Some designing product interview questions will involve the company's product or another company's product. When the question involves another product, understand what does this product does and how does it relate to your problem? For example, Microsoft asked how Twitter feeds can be summarized. Ask questions about how Twitter feeds are used. Is this going to be used to see user's reviews on new products or something else? Depending on how the product will be used, the goals will change. Further on the Microsoft-Twitter example, if Microsoft wants to see how people feel about new products/features, you would want to lean more towards a sentiment analysis than an actual summary of the text.

Once you have collected all the information required, you can start to design the product. At this point, tell the interviewer you would like some time to devise a design for the product. When you have a plan on how you are going to design your product, explain the steps thoroughly to the interviewer.

When collecting data/metrics, explain what data/where you will be collecting this data (if it is from an external source), and why you chose the certain data and user audience.

For example, Twitter asked, "How would you quantify the influence of a Twitter user?"

- **Who:** Who's data will you be collecting? Are you going to collect any users, verified users, users who are professional influencers? Explain why you are choosing this user group instead of another user group.
- **What:** What data will you be collecting? Are you collecting data about how many followers a user has? How many Twitter users view a certain influencing user's tweet in a day? Break down to the specifics of what data you will be collecting and the exact

purpose of choosing this data and how it relates to your goal.

- **Where:** In this example, Twitter data about a user will be in Twitter servers, so does not have to be mentioned. If you are planning on collecting data about a Twitter user outside of Twitter mention from which exact data/metric source you will be collecting from? Mention why you are collecting data from this company instead of any other source. If this is the only source to collect the data, mention that!

When mentioning the important features, explain how these features help the goal of this product. Continuing with the Twitter example: A possible important feature will be counting how many users click on a purchasing link that was posted by the influencing user. An increase in the number of users clicking on this link leads to an increase in the quantified influencing power of an influencing user.

Pain points are very important here as well. When designing any product, there will always be a pain point for a certain user demographic. Mention where there will be pain points in this product design and how you will tackle this problem. Is there any possible solution that can be implemented within the product or can any external implementation be used?

Now you have designed your product solution, but you have to measure how well the product will perform. At this point, you should **use the same framework for the Measuring Success question type**.

Certain designing product questions may go a step further and ask if a product should be released. The interviewee should think of how to gauge the potential success of the product.

- If there is a strong possibility of positive change in metrics and no negative impacts on other metrics in the short and long term, then we can deduce the product is probably going to be a success.
Example: Organic growth seems to be growing at a steady rate of 0.2% every month without major loss in DAU.
- If there is a negative impact on another metric (trade-off), find solutions to this negative impact. If it is not feasible to negate the negative effects, figure out which metric is more important to the company. As mentioned before, a general rule of thumb is, when the metric of the tradeoff is involved in the question, favor the metric in the question.
- If there are any long-term problems, understand if this is a solvable problem under projected circumstances, or else do further development of the product, so it doesn't cause unsolvable long-term problems.
Example: Major increase in product demand, causing the major supply shortage.

Now that you have gone through the initial general framework and the specific question framework, you have to wrap up your answer with the closing framework which should be applied to all 3 types of product interview questions.

Once you've given your specific answer to the question type, remember that your answer provides an overall solution to most use-cases. You should mention what are some edge cases and **mention how you will solve this edge case** using your solution. **Even if you don't know an answer to the edge cases, it is better to recognize the edge cases than not recognize them at all.** Providing the edge cases shows that you understand the product/feature well enough to realize these are edge cases. Don't try to combine your edge case into your main solution, since this may confuse the interviewer or yourself! Explain it after your main solution, so the interviewer knows you are talking about edge cases and also you could focus better on your overall solution.

Example: **Lyft asked** "Describe how to engineer the heatmap telling taxi drivers where to go to maximize their probability to get a client. How do you define which area will have high demand next and who do you want to go there?". Suppose your answer is to predict rides based on past data by hour, day of the week, month, and year. An edge case for this answer would be when there's a major event in the city, such as music festivals. Music festivals aren't always held at the same place during the same dates, so it could increase the

demand for Lyft rides exponentially for a couple of days then die down immediately, which is hard to predict if you just use past data. Real-world data collection of major events like these need to be incorporated.

After giving a systematic detailed solution to the answer, you should **provide a summary** to the interviewer.

Remember to include the applicable parts in your summary.

1. What are the metrics used in the solution and how it ties into your goal
2. Most likely cause of metric change and how to fix the change
3. Most important metrics for measuring success and when can the company conclude the product was a success
4. Most important parts of the design of the product

This summary is a key part of your product sense interview. It shows that you know how to explain your solution with the important parts. This trait is directly translated into a data science job. Imagine you are giving a rundown summary of why the problem arises and your solution to a client on why their product is failing.

Overview of steps to follow

1. Understanding question
 1. Clarify scenario/metric
 2. Product/Feature related
 3. Breaking down keywords from the question
 4. Time frame
2. Understanding internal/external influencing factors
3. Specific question-based framework
4. Edge cases
5. Summary of your solution

Communicating Your Solution

Your answer is not the only thing an interviewer looks for! They want to see how well you can communicate your thought process. This is a framework for how to communicate your solution.

A key part of your solution is to understand what the interviewer says. When communicating your solution remember to receive feedback from the interviewer. **Never follow the framework strictly.** Always keep the interviewer's comments in mind and change the steps/communication of your solution. Remember at the end of the day the interviewer will influence the decision if you are a good fit for the company or not.

Collect your ideas before answering the question

- Interviewees are allowed to take time to collect their thoughts. You do not want to end up halfway through your solution and realize you were going about the solution in the wrong perspective or you found a more accurate/efficient solution. Interviewers

expect you to take time before answering since this shows that you have thought through your solution and you are giving your best solution. Generally, people take around 30 seconds for these questions.

- If you are going to take a lot more time, maybe state the goals and some assumptions you are making about the question and ask if the goal/assumptions are accurate, so it does not seem like you are taking too much time to answer the question.

Agree upon goals/assumption of the question

- Deduce the goals of the question and make sure you and the interviewer are on the same page about the goals you are making about the question. These questions are usually ambiguous with their phrasing, so the interviewer can understand your unique perspective. This is the reason it is more important to communicate your thoughts than your final solution.
- The **assumptions you make in the beginning build the base of your solution, while the goal should show you how to go about your solution**. Example: [Twitter asked](#), "How would you quantify the influence of a Twitter user?". You could say "These are the assumptions I'm making. Assumption: There is a database that contains the tweets of users, once they're verified or when they have a substantial number of followers. The database will also include statistics about the user, such as the number of followers and the date-time a follower followed the user (The assumptions you make would change based on your solution). Are these plausible assumptions?"

For questions that ask how to improve a certain product, mention multiple target audiences but choose one for a solution

- Mention the three different target audiences who use this product/feature.
- Mention a pain point for each target audience
- Further, discuss why you have chosen one target audience and their pain point over the other two.
- Example: Exponent had a video on [how to improve Google Maps](#) (1:57 - 4:43)

Mention technical terms

- Technical terms should be used in your solution, but never force a technical term. You are applying for a technical role so you should show that you know important technical concepts and where they are directly applied to a business problem.
There are three instances where you should not use a technical term
 - You do not understand the technical term
 - If the technical term is an overkill to be used and there is a simpler solution
 - There is no relevance between the technical term and the goal/solution
- Sometimes company's mention certain technical terms on job descriptions, if certain technical terms are applicable during your interview questions, it is a good idea to mention them. For example, Facebook had a job description for a Data Science - Product Analytics position with "Applied statistics or experimentation (i.e. A/B testing) in an industry setting". A/B testing would be the technical term that could be possibly be used in your solution

If any step involves a new metric/variable, mention the following:

- Relevance to your current step in the solution and the goal
- How the metric/variable is derived. Example: If you are collecting CTR (Click through rate) for an Instagram ad, mention you will derive the CTR metric by calculating the number of times the ad is clicked divided by the number of times the ad is shown.

Always communicate your thoughts out loud! Talk out loud for each step, make sure each step is directly related to the previous step while also having a clear route to the goal.

If you ever feel you are going off track and your solution is not plausible, mention an instance where your product/feature would be thrown off. Take some time to see which step of your solution was closest to the goal and start a new path to the goal. Do not worry if you mess up! Everyone stumbles in their solution even during work, but it is a positive sign if you mention why your solution would not work rather than stick with a solution with a major flaw. Remember when you mention your new solution, how this solution will avoid the problem faced by the previously proposed solution.

After a couple of steps or if any important step was crossed, **ask the interviewer if all the steps make sense so far** and if you are heading in the right direction.

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